Research Report

Executive Summary

The global climate framework, anchored by the UNFCCC and Paris Agreement, faces a significant gap as current Nationally Determined Contributions track toward a 2.5°C temperature rise by 2100. Despite net-zero targets from major economies and the development of Carbon Capture, Utilization, and Storage (CCUS) technologies, global energy demand and temperatures continue to surge. Addressing these challenges requires overcoming the high energy requirements and operational risks of CCUS, alongside utilizing advanced predictive AI modeling to navigate updated climate scenarios.

Detailed Analysis

Global climate efforts are led by key stakeholders including the UN, IPCC, G20, and EU, with countries like Japan and the EU targeting net-zero by 2050, and China by 2060. However, current policies are insufficient, and the summer of 2023 was recorded as the hottest ever. To mitigate emissions, CCUS technologies capture CO_2 via pre-, post-, and oxy-fuel combustion. Point-source capture and Direct Air Capture (DAC) are primary methods, but DAC faces a net-carbon challenge where high-carbon energy use can cause emissions to exceed captured CO_2. Additionally, storage reservoirs like deep saline, basalt, and shale formations remain largely unexamined, and large-scale CCS implementation lacks drivers without emission-reduction incentives. To project future impacts, researchers use AI and machine learning to analyze atmospheric data and emissions under seven updated Shared Socioeconomic Pathways (SSPs) for the IPCC's 7th Assessment Report. If CO_2 levels stop increasing after 2050, warming may be limited to 1–1.5°C; otherwise, rising temperatures will continue to melt ice sheets, raise sea levels, and intensify tropical cyclones.

Key Takeaways

• Current Nationally Determined Contributions (NDCs) are insufficient to meet Paris Agreement targets, tracking toward a 2.5°C temperature rise by 2100.

• Direct Air Capture (DAC) faces a critical net-carbon challenge where its operational emissions can exceed captured CO_2 if powered by high-carbon energy.

• Geological storage options like deep saline, basalt, and shale formations remain largely unexamined, and CCS lacks drivers without emission-reduction incentives.

• The IPCC's 7th Assessment Report utilizes seven streamlined Shared Socioeconomic Pathways (SSPs) that exclude previous worst-case and best-case projections.

• Predictive artificial intelligence and machine learning models are increasingly utilized to analyze historical data, atmospheric conditions, and emissions to forecast complex climate scenarios.

Conclusion

Achieving global climate targets requires bridging the gap between current policy commitments and actual emission reductions. While CCUS and advanced AI-driven predictive modeling offer critical pathways for mitigation and analysis, their success depends on overcoming technical, energy, and storage constraints, alongside establishing stronger policy incentives to drive large-scale implementation.